



There are Plumes on Europa

Old data from the Galileo probe analysed by new simulations shows it flew through a plume of water on Europa. <https://digit.in/jun18-A60-1>



Gotham renewed

Gotham has been renewed by Fox for Season 5, however, it will be the final season to air. <https://digit.in/jun18-A60-2>

Personalised peripherals

Fancy getting yourself a bespoke keyboard and mouse? Look no further.

Mithun Mohandas | mithun@digit.in

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henever we suggest keyboards and mice to our readers, we always make it a point to mention that

they are both very personal devices and you should only finalise your choice of keys and keycaps once you've tried them all. And grips when we're talking about mice. That's why when you go to a store, you'll always find a key tester kit with all the different mechanical keys placed in serial order. You can figure out what kind of travel distance or actuation force feels most comfortable for you as you go through all the switch options. However, of late, it seems that most folks don't like what mainstream manufacturers have to offer. So what can you do to appease the whims of your fingers? Build your own keyboard of course!

KEYBOARD

The first step towards building your very own customised keyboard would be to go try out all the variants first. And we mean variants of both, the mechanical keys as well as the sizes of keyboards. Yes, they both have variants and if that was surprising for you, then here's a small list of the different options for keys/switches, keycaps, keyboard materials and keyboard sizes.

Mechanical Key types:

- Cherry MX - Black, Silent Black, Brown, Blue, Red, Green, Silent Red, Clear, White, Nature White, Tactile Grey, Linear Grey, Silver
- KBT - Black, Brown, Blue, Red
- Greentech - Black, Brown, Blue, Red
- Gateron - Black, Brown, Blue, Red, Clear, Green
- Matias - Click, Quick Click, Linear
- Topre - 30G, 35G, 45G, 55G, Variable
- Kailh - Black, Brown, Blue, Red, Green

Mechanical Keyboard sizes:

- 20%, 40%, 60%, 65%, 75%, TKL, 80%, 95%, Full-Frame.

Keyboard materials:

- Aluminium, Plastic, Wood

Key caps:

- Single plastic, double shot,

What we've mentioned here are just the most easily customisable parts that you can buy. You can even go further but that's for another day. Remember that buying these pieces one-by-one does jack the cost up significantly.

To begin with, you'll need to get a PCB and a case/frame for your new keyboard. You can either get a PCB that supports hot-swapping or you can go the more durable way and solder each key onto the PCB. So yeah, you'll need to polish your soldering skills. The key parameters you need to keep in mind while picking the PCB is the size that you are most comfortable with. Almost all of the PCBs have a similar functionality set i.e. you will get all the

numbers and alphabets. A full-frame keyboard has everything including the letter block, arrow keys, navigation keys, function keys and the numpad. With TKL, you lose out on the numpad and then if you ditch the navigation keys and the arrow key cluster, you're left with a 60% keyboard. Remove the function keys and you've got a 40% keyboard. As the size grows smaller, you lose out on functionality but gain portability. Also, the full frame keyboard becomes way more expensive because of all the additional switches that you'll have to buy. Let's pick the 60% layout which hits the sweet spot between size and functionality.

The PCBs come with all the circuitry pre-soldered. So you don't have to worry about soldering micro-pitch wires. All you need to worry about are the individual switches which have very thick leads. The PCB we've picked is a GK64 60% RGB that supports switch hot-swapping. If you don't want hot-swap capability then you can always check out the DZ60 which offers a few variants within the 60% layout.



GK64 RGB Hot Swap PCB

Once that's done, you can pick the case/frame for your keyboard or you can make one. You only need to keep in mind where the alignment studs are. If you plan to buy them, then plastic cases come for about \$15 for the 60% layout. You can opt for opaque, clear, frosted plastic with varying textured finishes. Anodised aluminium starts at \$11 and wood is the most expensive at \$69 being the starting price. All you need to ensure is that the case is compatible with the PCB i.e. the cable interface lines up with the slot on the case.